

# INTERNATIONAL news

## US to announce billions in chip subsidies for construction of new factories

The US plans to announce multi-billion dollar subsidies for chip companies, including Intel, TSMC, and Samsung Electronics, to support the construction of new semiconductor factories in the US and boost the manufacturing of advanced semiconductors used in smartphones, artificial intelligence, and weapons systems. Initial funding awards have already started, with over \$35 million granted to a BAE Systems facility for chip production for fighter planes.

## China turns to open source RISC-V amid rising US restrictions

To counter US sanctions and reshape the global semiconductor industry, China is increasing its reliance on geopolitically neutral open source instruction set architecture RISC-V for high-performance chips, according to the observable trends. In China, at least \$50 million have been invested in projects involving RISC-V between 2018 and 2023. The People's Liberation Army's (PLA) Academy of Military Sciences has been at the forefront.

## Synopsys to acquire simulation software company Ansys

Synopsys is set to acquire Ansys in a \$35 billion cash-and-stock deal, marking the largest technology sector acquisition since Broadcom's takeover of VMware. This combines Synopsys' electronic design automation (EDA) tools with Ansys' simulation and analysis portfolio. If the deal falls through, Synopsys will owe Ansys a \$1.5 billion termination fee, while Ansys would pay \$950 million if it accepts a superior proposal.

## Renesas to acquire electronics design firm Altium

Japan's Renesas Electronics plans to acquire electronics design firm Altium for \$5.9 billion in cash. Renesas, supplier of chips to major automakers like Toyota and Nissan, aims to provide device design through Altium's digital tools for engineers and designers working on circuit boards. The transaction has received approval from both companies' boards but still requires approval from Altium shareholders, an Australian court, and regulators.

## Infineon and GlobalFoundries join to drive automotive Innovation

Infineon Technologies and GlobalFoundries have entered into a strategic multi-year agreement to increase the supply of Infineon's 40 nanometre AURIX TC3x automotive microcontrollers, power management, and connectivity solutions from 2024 through 2030. A key element of this collaboration is the integration of a highly reliable embedded non-volatile memory (eNVM) technology solution, tailored for mission-critical applications in the automotive industry.

## OpenAI CEO Sam Altman to foray into chip manufacturing

OpenAI Chief Executive Officer (CEO) Sam Altman aims to set up a network of semiconductor manufacturing plants. Altman is in discussions with middle eastern investors and chip fabricators, including TSMC, about launching the venture. This could potentially require up to \$7 trillion in funding. Current chip manufacturing capacity is unable to keep up with swift AI deployment.

## Nvidia and Equinix provide AI supercomputers to businesses

Equinix, a data centre operator, and chipmaker Nvidia announced their partnership to provide Nvidia's supercomputing systems to business clients. As part of the agreement, Nvidia has equipped Equinix staff with the necessary skills to construct and manage its systems. Businesses will buy Nvidia's systems but will rely on

Equinix for efficient operation and maintenance, while maintaining ownership of the systems.

## SK hynix, NTT, Intel join hands to develop optical chips

NTT will partner with Intel for CPUs and with SK Hynix for memory chips to incorporate light-receiving devices within chips by 2027 and advance memory technology capable of storing terabit-level data. Intel will reduce power consumption by 30-40%. NTT will improve power efficiency through ion technology, with the goal of increasing it by tenfold by 2025 and up to 100 times by 2032. The Japanese government is providing \$304 million to advance next-generation 6G communication research.

## Nokia and Dell collaborate to deploy 5G networks

Nokia and Dell Technologies will partner to deploy private 5G networks and adapt networks to the cloud. Customers of Nokia AirFrame, the telecom equipment maker's cloud-focused data centre business, will eventually transition to Dell's PowerEdge servers, which are specifically designed for modern telecom network workloads. The companies plan to build on their existing partnership and work towards integrating NDAC with Dell's NativeEdge software platform.

## UAE fund to support growth of advanced technology in developing nations

The UAE launched a \$200 million fund to support the growth of advanced technology in developing countries. The fund, backed by the Abu Dhabi government and managed by the UAE's Advanced Technology Research Council (ATRC), aims to catalyse the development and promotion of sophisticated technologies to assist governments and organisations in less affluent nations. Technologies developed within the Gulf state will be shared with developing nations with required support for research projects tailored to their specific needs.

## Nokia and Vivo enter 5G patent licence agreement

Nokia has entered a multi-year 5G patent licence agreement with Chinese smartphone manufacturer Vivo. Nokia's intellectual property licensing division, Nokia Technologies, is expected to generate a minimum operating profit of \$1.51 billion in 2024. This deal marks Nokia's sixth major smartphone licensing agreement in the past thirteen months, nearly completing its cycle of smartphone licence renewals.

## Global symposium for quantum technology implications set for March

A Global Quantum Symposium will take place from March 18-20, 2024. It will bring together over 40 leaders from academia, industry, and governance to discuss hardware and software aspects of quantum technology, as well as its ethical, political, and career implications. The event will be held at the campus of uptown Basel in Switzerland, near QuantumBasel, a centre dedicated to fostering collaboration in quantum and AI technologies.

## Toppan Photomask and IBM agree for joint R&D on semiconductor EUV photomasks

IBM and Toppan Photomask have signed an agreement to collaborate for five years, starting from the first quarter of 2024, on developing two nanometre (2nm) logic semiconductor node technology using extreme ultraviolet (EUV) lithography. It includes the development of High-NA EUV photomask capabilities. The partnership will focus on improving photomask capabilities at both the Albany NanoTech Complex in Albany, NY, USA, and Toppan Photomask's Asaka Plant in Niiza, Japan.